

Planning to Action: A Field Experiment on Soft Commitment and Academic Performance

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ABSTRACT

Procrastination affects up to 80% of university students, often hindering academic performance. To tackle this, we conducted a pre-registered field experiment with over 2,000 students at a German university, randomly assigning them to one of three groups: a semester-long online learning journal, the same journal coupled with a soft commitment device, or a control group. Analysis of administrative data reveals that the journal and commitment device group achieved significantly higher earned credits and GPA (standardized effect sizes of 0.075 and 0.12, respectively). In contrast, the journal-only group did not experience notable gains. These results highlight the importance of integrating commitment mechanisms into low-touch interventions, as they enhance accountability and maximize overall effectiveness. Further investigation into the mechanisms suggests a dual effect of the intervention: low-procrastinators experience long-term academic benefits, while high-procrastinators show short-term improvements in task initiation.

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I. Introduction

Despite the substantial wage premiums associated with higher education, university completion rates remain inefficiently low. Across OECD countries, around 40 percent of students graduate within the official duration of their program (OECD, 2025). A growing literature suggests that these outcomes are driven not only by financial or institutional constraints but also by behavioral frictions, specifically, failures in self-regulation. Procrastination is widespread among university students: over 80 percent report delaying academic tasks, and roughly half do so habitually in ways that harm performance (Steel, 2007; Klingsieck, 2013). These tendencies are particularly pronounced during the transition to higher education, when external structure is limited, and the demand for self-regulation increases sharply (Vosniadou, 2020).

Policy responses to these behavioral barriers face a fundamental trade-off between effectiveness and scalability. Resource-intensive interventions, such as individualized therapy or coaching, can effectively reduce procrastination but are difficult to scale across large university cohorts (Rozenal et al., 2018). Conversely, scalable "low-touch" interventions, such as informational nudges, are inexpensive but often lack the intensity required to alter complex, repeated behaviors like studying (Damgaard and Nielsen, 2020). Behavioral tools such as commitment devices occupy a promising middle ground, having been shown to effectively increase student effort and credit accumulation (Himmeler et al., 2019). Yet, despite this potential, evidence on their efficacy in fully scalable, low-cost formats remains limited (Gallego et al., 2023).

This paper reports results from a pre-registered⁶ field experiment testing whether a low-cost, scalable intervention can improve academic performance among university students. We randomized a complete cohort of over 2,000 incoming students at a German university into three groups. The control group received a placebo letter. The first treatment group received access to a semester-long online learning journal based on the Mental Contrasting with Implementation Intentions (MCII) framework (Oettingen and Gollwitzer, 2010), which structures weekly goal setting, reflection, and planning. The second treatment group received the same journal supplemented with a soft commitment device: a voluntary target agreement⁷ through which participants could formally commit to their self-defined study goals. Soft commitment devices impose psychological costs for non-compliance, thereby strengthening adherence to planned actions without material enforcement (Bryan, Karlan & Nelson, 2010). This design allows us to isolate the incremental effect of adding a commitment mechanism to an otherwise identical planning tool, directly testing whether planning interventions require accountability to be effective.

We find that students assigned to the journal-with-commitment group achieved significantly better academic outcomes by the end of the first semester. Specifically, this group earned 0.85 additional course credits (0.075 SD; $p < 0.10$) and improved their GPA by 0.12 standard deviations

⁶ Link to the pre-registration: <https://www.socialscienceregistry.org/trials/10147>

⁷ We inform the students that the target agreement is voluntary and carries no consequences if they fail to meet the goals, positioning it as a 'soft commitment device' as discussed by Bryan et al. (2010).

($p < 0.05$) relative to the control group. The simultaneous improvement along both dimensions rules out a quantity-quality trade-off, ensuring that students did not sacrifice grades to accumulate more credits⁸. In contrast, offering the learning journal without the commitment device yielded no significant effects on any outcome. This sharp contrast underscores the importance of accountability in low-touch interventions: while the journal's planning components alone were insufficient to alter behavior, pairing them with a mechanism that raises the psychological cost of deviation produced meaningful gains.

In both treatment groups, approximately 50% of students accessed the treatment link at least once, while 18% in the second group opted to sign the commitment device. Since not all assigned students engaged with the learning journal, the intention to treat (ITT) estimates represent a lower bound for the intervention's average treatment effect on academic achievement. To better understand the impact of actively using the learning journal, beyond simply offering it, we estimated treatment-on-the-treated (TOT) effects. We defined treatment take-up as any student who opened the intervention link at least once, reflecting engagement with the treatment. The TOT analysis shows stronger effects in the second treatment group, where treated students earned 1.7 additional credits (0.15 SD, $p: 0.07$) and had better GPA (0.22 SD, $p < 0.01$), indicating a small to moderate treatment effect.

We expect treatment effects to vary depending on the degree to which students benefit from external structure. Prior literature demonstrates that individuals with present-biased tendencies benefit disproportionately from calendar-based goal setting (Grohmann, Lakemann, and Seitz, 2020) and commitment devices (Himmler et al., 2019). To test this, we investigate pre-registered heterogeneous treatment effects based on students' procrastination tendencies⁹. The results reveal a clear divergence: the commitment treatment improves academic outcomes for both high- and low-procrastinators, but along different margins.

For high-procrastinators, the intervention primarily affects the extensive margin of task initiation. These students earn 1.18 additional credits ($p < 0.10$) under the commitment treatment, with no corresponding change in GPA. Evidence on mechanisms suggests that the commitment device counteracts delays at a critical early juncture. High-procrastinators report significantly lower procrastination tendencies during the course registration period (0.33 SD; $p < 0.05$) and sign up for 1.7 more credit points ($p < 0.05$) than comparable students in the journal-only group. Because course registration is a discrete, deadline-driven task early in the semester, this pattern is consistent with the commitment device facilitating timely task initiation, thereby increasing subsequent credit accumulation.

For low-procrastinators, who are less constrained by task initiation frictions, the intervention operates along the intensive margin of academic performance. These students experience a significant improvement in GPA (0.138 grade points; $p < 0.01$), with no increase in credit volume.

⁸ Enhancing student educational outcomes early in their academic journey is particularly important, as it can set a positive trajectory towards degree completion, as evidenced by the concept of academic momentum (Attewell et al., 2012).

⁹ We use students' university application and matriculation dates as a proxy for procrastination, assuming that late applicants are more likely to procrastinate than early ones. This approach is widely used in the literature (e.g., De Paola & Scoppa, 2015; Reuben et al., 2015; Himmler et al., 2019).

Consistent with this interpretation, low-procrastinators are more likely to take up the voluntary commitment device (20% versus 15.8%), suggesting that they use it to support sustained self-regulation rather than to overcome starting inertia. In combination with the journal’s reflective and planning components, this pattern points to improvements in study quality that accumulate over the semester. Taken together, the heterogeneity and mechanism evidence suggest that the commitment device operates through multiple channels, addressing task initiation for some students and sustained effort for others.

The remainder of the paper is structured as follows. We begin by reviewing the literature and presenting the conceptual framework and background information of the study. Section II explores procrastination behavior, its impact on study performance, and intervention take-up. Section III outlines the empirical strategy and examines the treatment effects, including ITT and TOT estimates, heterogeneous effects based on procrastination, and underlying mechanisms. Finally, Section IV concludes.

A. Related Literature

This study introduces a novel approach by combining schedule assistance with a soft commitment device aimed at reinforcing students’ dedication to their learning inputs, a combination not previously explored in the literature. Existing research demonstrates that scheduling tools alone often fail to improve academic outcomes. For instance, Oreopoulos et al. (2019) provided university students with an online planning tool to create semester-long study schedules and sent periodic reminders, yet found no significant effects on grades or credit attainment—a result mirrored in our first treatment group. Similarly, Li and Baker (2018) observed that email prompts encouraging students to schedule lecture video viewing in an online course had no impact on performance. Even outside education, Grohmann et al. (2020) reported that offering a savings calendar to small business owners did not increase savings, despite high demand among present-biased individuals. These findings collectively suggest that scheduling interventions, when implemented in isolation, lack the ability to drive behavioral change. Our study addresses this by integrating a commitment device with scheduling support, a combination we test directly in our experimental design.

Our intervention’s focus on study-time goals further contributes to the goal-setting literature, which debates the efficacy of task-based versus performance-based goals. Clark et al. (2020) found that students who set weekly task-based goals improved their course performance, while end-of-semester grade goals (performance-based) had no effect. Conversely, Van Lent and Souverijn (2020) demonstrated that performance-based goals set during one-on-one counseling sessions significantly boosted final grades. This discrepancy in the results may stem from differences in the commitment levels of the students towards their goals: Van Lent and Souverijn’s (2020) mentorship setting likely fostered stronger dedication to goals compared to Clark et al.’s (2020) classroom-based approach. Our findings align with this interpretation, underscoring the importance of embedding commitment mechanisms into goal-setting frameworks.

Finally, our results advance the understanding of “soft” commitment devices in low-touch educational interventions. Commitment strategies have successfully changed behaviors in diverse contexts, such as exercise adherence (Royer et al., 2015), savings (Gugerty, 2007), and smoking cessation (Giné et al., 2010). In education, Himmler et al. (2019) showed that voluntary study-plan commitments increased credit attainment, particularly among procrastinators. Similarly, Felkey et al. (2021) found that microcommitments raised exam grades by 3.5 percentage points in online courses, with stronger effects for procrastinators. Patterson (2018) further demonstrated that commitment devices limiting internet distractions improved study time and grades by 0.29 standard deviations, whereas reminders or blockers alone had no impact. Our study reinforces this pattern: the commitment component of our intervention drives behavioral change more effectively than journaling alone, highlighting its value as a scalable, low-cost tool to enhance academic success.

B. Conceptual Framework

Procrastination in higher education reflects a confluence of behavioral frictions rather than a single failure mode. Three mechanisms are particularly well-documented. First, self-regulation deficits: students who struggle with impulse control and delayed gratification are systematically more likely to postpone academic tasks (Wijaya & Tori, 2018; Steel, 2007). Second, present bias and time inconsistency: when the costs of effort are immediate, but the rewards are distant, quasi-hyperbolic discounting leads to repeated postponement of tasks that individuals themselves plan to complete (O'Donoghue & Rabin, 1999; Augenblick et al., 2015). Third, weak goal commitment: vague objectives reduce the psychological cost of deviation, making it easier to abandon study plans in the face of competing demands (Flett et al., 2012; Koch & Nafziger, 2011). While each channel is individually established, the challenge for intervention design is to address them jointly within a scalable format.

Our intervention targets these channels through two complementary components. The learning journal provides a structured weekly cycle of goal setting, self-monitoring, and reflection. Students set study-time targets for the upcoming week and receive automated feedback comparing these targets to their self-reported hours. This design follows the MCII framework (Oettingen & Gollwitzer, 2010), which strengthens self-regulation by prompting individuals to form concrete “if-then” plans (Duckworth et al., 2013; Kizilcec & Cohen, 2017). By mandating specific goals and regular self-assessment, the journal directly addresses self-regulation and goal-clarity deficits.

However, planning tools without accountability mechanisms often yield null effects because they reduce ambiguity without raising the cost of deviation (Oreopoulos et al., 2019; Grohmann et al., 2020). Our second component, a soft commitment device, addresses this gap by offering students the opportunity to sign a target agreement. Although the agreement carries no formal sanctions, it imposes a psychological cost of noncompliance (a broken self-promise) that raises the salience of planned actions relative to immediate temptations (Bryan et al., 2010). This mechanism counteracts present bias by shifting the perceived cost of deviation forward in time.

The main challenges that face our intervention are take-up and long-term persistence. Although students with self-control issues theoretically seek support¹⁰, the reality is more complex. In our case, 77% of students are first-time university attendees, making them more likely to be unaware of their time-inconsistent tendencies or lacking the experience to recognize the intervention's benefits¹¹. Maintaining engagement with the intervention is challenging, as students may lose focus or forget tasks over time (Lei et al., 2011; Ericson, 2017). To address this, we send weekly reminders to prompt students to complete their learning journals, helping refocus attention and keep the intervention salient as they adapt to the academic environment (Guynn et al., 1998; Ericson, 2017).

C. Institutional Background

We conducted our intervention at a large public University of Applied Sciences (UAS) in Germany. The field experiment included the entire cohort of 2,221 incoming first-semester students who enrolled in one of 21 bachelor's programs in the winter semester of 2022-23. The curriculum at UAS is generally more practice-oriented than at traditional, more research-oriented German universities. The study programs at the UAS are structured in accordance with the European Credit Transfer and Accumulation System¹². A standard bachelor's degree comprises 210 credits, ideally completed within 7 academic semesters. However, the average time to degree is typically around 8.5 semesters. The standard advice according to the study plan is to earn 30 credits per semester, which is equivalent to 15 credits in the U.S. system. The grading system ranges from 1 to 4, with 1 being the highest grade achievable and 4 indicating the lowest passing grade, i.e., the lower the grade the better.

UAS serve a substantial and growing share of the German student population. For instance, in the fall semester of 2020, about 40% of freshman students started studying at a UAS (Statistisches Bundesamt, 2024). Our study features a sample with observable characteristics that are very similar to the general student population in Germany. Approximately 53% of our sample hold a degree from the highest secondary education track—the so-called “Abitur”, mirroring the 53.5% among freshmen at all German UAS in 2020¹³. Additionally, the average high school GPA of our sample is 2.56, aligning closely with the 2.41 GPA of all German high school graduates in 2019¹⁴.

Higher education, especially in Germany, promotes a significant level of autonomy and independence for its students. German universities traditionally follow a less rigid structure, with fewer enforced deadlines, less feedback through continuous assessments like midterm exams, and more self-directed learning (Bäulke & Dresel, 2023; Stallmann, 2002). This academic freedom and increased responsibility for self-management can pose challenges in time management and

¹⁰ For example, several studies have demonstrated the link between commitment contract demand and time-inconsistent behavior (Augenblick et al., 2015; Kaur et al., 2015; Houser et al., 2018).

¹¹ Previous research, such as Kaur et al. (2015), has shown that take-up of commitment contracts is positively correlated with indicators of time inconsistency, but this effect is more pronounced after repeated exposure to the contracts. This suggests that learning from repeated decision-making plays a critical role in understanding the demand for such interventions.

¹² Europe-wide, universities use a standardized point system (European Credit Transfer and Accumulation System, ECTS), under which a semester consists of 30 credits, with a typical workload of 25-30 hours per credit. (see: <https://education.ec.europa.eu/education-levels/higher-education/inclusive-and-connected-higher-education/european-credit-transfer-and-accumulation-system>)

¹³ See: <https://www.datenportal.bmbf.de/portal/en/Tabelle-2.5.106.html#A1>

¹⁴ See: <https://www.kmk.org/dokumentation-statistik/statistik/schulstatistik/>

contribute to procrastination (Dietz et al., 2007; Hen & Goroshit, 2018; Bäumke & Dresel, 2023). Additionally, since tuition fees at most German universities are heavily subsidized, financial pressures are minimal. This may reduce incentives to maintain a rigorous credit load, potentially exacerbating procrastination, as staying matriculated at a university comes with perks¹⁵.

D. Field Experiment

At the beginning of the winter semester of 2022-23, students were randomized into two treatment groups and a control group (see Table 1 for an overview of the experimental design). Students in the treatment groups received an unannounced letter that introduced the intervention, while the control group received a placebo welcome letter (see Section F of the Appendix). During the semester, the students in both treatment groups received another letter with a reminder of the learning journal (see Figure 1 for an outline of the intervention).

Treatment 1: learning journal.—Students in the first treatment group were offered access to a semester-long online learning journal. The journal enables students to reflect on their study behavior from the previous week and set goals for the upcoming week, allowing them to compare intended study hours with actual time spent. At the beginning of every week in the semester, the journal encourages students to:

1. Report on their daily study hours from the previous week, with notifications (from the second week onward) comparing actual study time to initial goals.
2. Set study time goals for the upcoming week.
3. Reflect on positive outcomes of achieving their goals, identify obstacles, and develop strategies to overcome them.

This design leverages aspects of MCII interventions, shown to enhance self-regulation, goal commitment, and goal-directed behavior (Duckworth et al., 2013; Kizilcec & Cohen, 2017). Both treatment groups received weekly email reminders to fill out the journal. A sample of the journal for a typical week, along with the invitation letters, is provided in Section F of the Appendix.

Treatment 2: learning journal + commitment.—The second treatment group received the same learning journal, with the addition of a soft commitment device. This device allowed students to voluntarily commit to their study time goals through a target agreement. Students were, in a short paragraph, informed of the potential benefits of a commitment device and provided with the opportunity to formally commit by signing their name under a target agreement at the beginning of the learning journal. The agreement specifically commits the students to the study time goals that they set for themselves, while informing them that signing the agreement is voluntary and that failing to meet the goals has no consequences (see Section F of the Appendix). Students who chose to sign the target agreement were reminded of it in the form of a text box in the learning journal.

¹⁵ Students enjoy certain perks like subsidized housing, healthcare, and transportation, which can incentivize extending their studies.

TABLE 1—EXPERIMENTAL DESIGN

T0: Control	T1: Learning Journal	T2: Learning Journal + Soft Commitment
1. Welcome letter/email incl. information on: i) Typical workload of a full-time study program. ii) Announcement of two short online surveys that will be conducted during the semester. iii) Second page with information on workload according to ECTS, the competence center, a diversity statement of the university, and advisory services offered by the university. 3. Two short surveys on study behavior and well-being 4. Second letter/email incl. information on: i) Upcoming exams ii) (Updated) info on points i), ii), and iii) of the welcome letter.	1. Welcome letter/email incl. information on: i) Typical workload of a full-time study program. ii) Announcement of two short online surveys that will be conducted during the semester. iii) Second page with information on workload according to ECTS, the competence center, a diversity statement of the university, and advisory services offered by the university. <i>iv) Invitation to the online learning journal.</i> 2. Online learning journal incl.: <i>i) Weekly reminder email</i> 3. Two short surveys on study behavior and well-being 4. Second letter/email incl. information on: i) Upcoming exams ii) (Updated) info on points i), ii), and iii) of the welcome letter. <i>iii) Learning journal reminder</i>	1. Welcome letter/email incl. information on: i) Typical workload of a full-time study program. ii) Announcement of two short online surveys that will be conducted during the semester. iii) Second page with information on workload according to ECTS, the competence center, a diversity statement of the university, and advisory services offered by the university. <i>iv) Invitation to the online learning journal.</i> 2. Online learning journal incl.: <i>i) Weekly reminder email</i> <i>ii) Voluntary target agreement (soft commitment device)</i> 3. Two short surveys on study behavior and well-being 4. Second letter/email incl. information on: i) Upcoming exams ii) (Updated) info on points i), ii), and iii) of the welcome letter. <i>iii) Learning journal reminder</i>

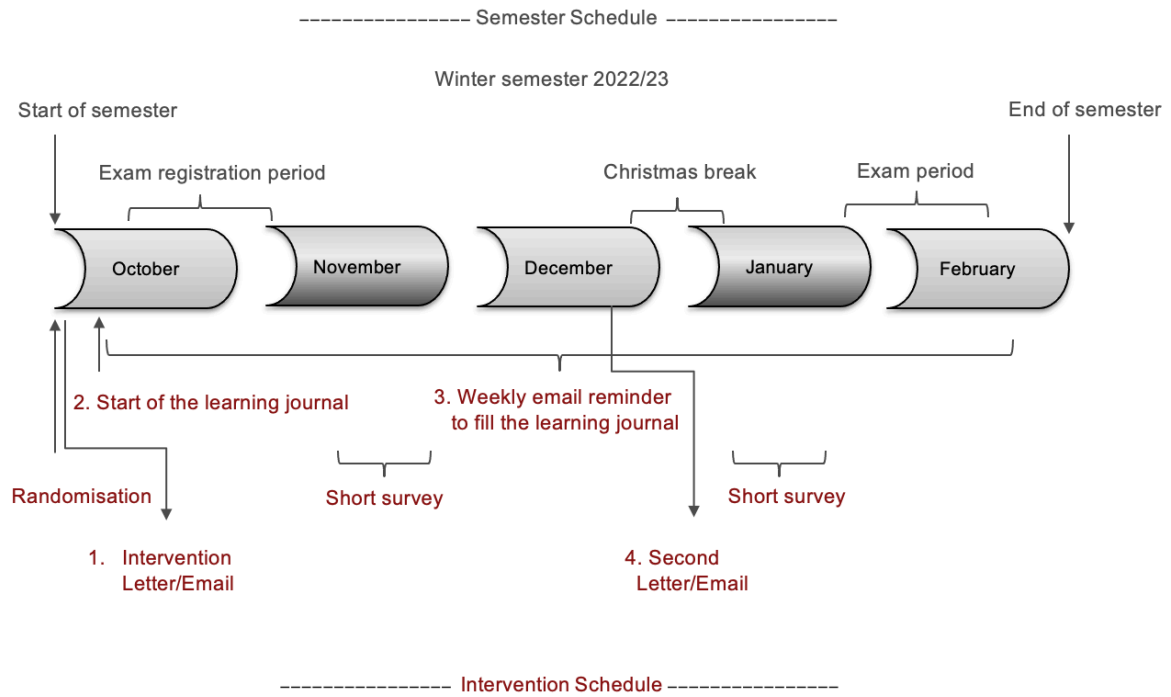


FIGURE 1. TIMELINE OF THE INTERVENTION AND UNIVERSITY SEMESTER

Notes: The figure shows the timeline of the intervention and the university semester in the winter term of the year 2022. The experiment began after all COVID-19 pandemic measures were removed. All classes and exams were held face-to-face.

Randomization.—We randomized all incoming students into three experimental groups. Randomization was carried out using a stratification and balancing procedure, where the strata were constructed based on the study program, a median split of the high school GPA distribution¹⁶, and a median split of students based on their procrastination index. This ensures a balanced treatment assignment across all study programs and important student characteristics. See Section D of the Appendix for further details on the randomization procedure.

To check whether our randomization procedure successfully balanced observed characteristics across the three experimental groups, we provide estimates of summary statistics in Table 2. Columns 1–3 report the means for each variable across the three groups. Column 4 provides p-Values for a joint orthogonality test. We find no significant differences in any of the students' characteristics. Furthermore, differences are quantitatively very small. At baseline, the average age of our study population was just over 21 years, with a 38% female share. On average, students enroll in university 1.5 years after their high school graduation date, with around 75% of them enrolling for the first time in a higher education institution.

¹⁶ If the study program had small number of students (under 24), we proceed without splitting the sample by high school GPA. See appendix section D for a detailed description.

TABLE 2—BASELINE CHARACTERISTICS & BALANCING

	Control	Journal Treatment	Journal + Commitment Treatment	Joint Orthogonality Test (p-value)	N
	(1)	(2)	(3)	(4)	(5)
<i>Socio-economic background</i>					
1[Female]	0.38	0.38	0.35	0.306	2221
Age at enrollment (years)	21.43	21.23	21.41	0.397	2221
1[A-Level]	0.53	0.53	0.51	0.802	2221
1[schooling outside of Bavaria]	0.11	0.10	0.08	0.273	2221
1[foreign schooling]	0.04	0.05	0.04	0.558	2221
<i>Study characteristics</i>					
Procrastination index (std.)	-0.001	-0.012	0.011	0.907	2221
Matriculation date (average)	Aug 11, 2022	Aug 11, 2022	Aug 10, 2022	0.890	2221
Application date (average)	Jun 23, 2022	Jun 23, 2022	Jun 24, 2022	0.644	2216
1[Missing application date imputed]	0.06	0.06	0.06	0.739	2221
High school GPA (German scale: 1 very good - 4 very poor)	2.47	2.47	2.49	0.785	2221
1[Missing high school GPA imputed]	0.02	0.02	0.02	0.818	2221
1[first study program]	0.77	0.76	0.75	0.625	2221
Time between school degree and enrollment (years)	1.68	1.56	1.62	0.624	2221
Number of students in each group	740	740	741		

Notes: The Table shows group means of main outcomes at baseline for the control group and two treatment groups. The final column reports the p-value of an F-test on the treatment arms. Variables in brackets with a leading digit 1 (1[...]) are binary variables. The last row reports the number of students in each group. Small differences in group sizes arise because randomization was conducted within strata blocks, with some strata blocks having block sizes that were not multiples of three (the number of randomization groups). Stratification variables include high school GPA and the procrastination index. Study characteristics of all students were collected before randomization.

E. Data Sets

To evaluate the effects of the intervention, we draw on a rich set of complementary data sources that capture both academic outcomes and underlying behavioral mechanisms. These sources include:

1. **Administrative Data:** University-provided administrative records were used for the randomization process and served as both covariates and outcome variables. This dataset includes students' background information, application dates, and academic performance metrics.
2. **Online Self-Assessments (OSAs):** Mandatory for students in nine study programs, these assessments were conducted before the start of the semester. The OSAs gathered information on time preferences, procrastination tendencies, and socio-economic background.
3. **Online Surveys:** Two voluntary online surveys were administered during the semester to collect data on study behavior and non-cognitive outcomes, including stress levels and student satisfaction.
4. **Learning Journal:** The learning journal provided insights into the usage patterns, weekly study hours, and academic goals of the treatment groups. It also measured the take-up rate of a soft commitment intervention for the second treatment group.

F. Outcome Variables

Study performance measures. As preregistered, the study's primary focus includes passed course credits, GPA¹⁷, and student persistence (dropout rate) at the first semester's end. To capture the

¹⁷ GPA is only computed for students who earned credits. Students with zero earned credits do not have a GPA, which explains the lower observation number in respective analysis.

overall academic achievement, we form an index using a standardized inverse-covariance weighted average (Schwab et al., 2020).

We use survey data to examine whether the treatment reduced procrastination tendencies post-intervention. Additionally, we assess potential side effects by analyzing its impact on student engagement, measured through credit points registered and attempted, as well as hours dedicated to studying. To ensure a comprehensive assessment, we evaluate the intervention’s influence on students’ well-being by examining non-cognitive outcomes collected in online surveys. See Section E of the Appendix for a detailed description of all outcome variables.

II. Procrastinating Behavior, Study Performance, and Intervention Take-up

A. Procrastinating Behavior

Procrastinators, by definition, are inclined to delay tasks, which includes postponing participation in our treatment program (O’Donoghue & Rabin, 1999). This delay can hinder timely access to the treatment, thereby reducing its potential positive impact on educational outcomes. Therefore, it is important to note that when targeting procrastination, a trade-off characterized by negative selection on gains might arise (Roy, 1951). Paradoxically, while students who procrastinate stand to benefit the most from our intervention, they are also less likely to engage in it. Following the Roy Model, this negative selection on gains introduces a self-selection bias leading to an underestimation of the intervention's efficacy among procrastinators.

From this understanding, we can derive several empirically testable predictions: students with higher procrastination tendencies are likely to have lower academic performance, a lower take-up rate of the intervention, and later engagement with the intervention. No clear prediction can be made regarding the expected treatment effects on educational outcomes among procrastinators. This is because the potentially higher gains for procrastinators from our treatment may be canceled out by lower and later treatment take-up due to procrastination tendencies.

To measure procrastination tendencies, we construct a continuous procrastination index based on students’ application and enrollment date, following the approach of De Paola and Scoppa (2015). This method relies on observed choices rather than self-reported data, mitigating concerns of systematic measurement error due to social desirability bias. In the German higher education system, the application period typically runs from early May to mid-July, followed by a registration window from August to the end of September. We define procrastination as the delay in completing these tasks: students who apply and enroll late exhibit higher procrastination tendencies, while those who do so early are considered low procrastinators. This approach assumes that early application reflects prompt task initiation, whereas later application signals procrastination. Thus, our measure captures procrastination on a continuous scale, where longer delays from the start of the application and enrollment period indicate stronger procrastination tendencies. We proceed by investigating the

predictive power of the procrastination measure on educational outcomes and the take-up and usage rate of the intervention.

B. Procrastination and Study Performance

Using our procrastination proxy, we calculated the median within each stratum from the randomization procedure and classified students as high-procrastinators if their value was above their stratum-specific median, and as low-procrastinators otherwise. In Table 3 we compared the academic performance of both groups. Aligning with a meta-analysis of 33 studies by Kim and Seo (2015), which found a negative correlation between procrastination and academic performance, our findings show that high-procrastinators consistently underperform relative to their low-procrastinating peers. On average, low-procrastinators earn 17.2 credits, maintain a GPA of 2.49, and have a 6.8% dropout rate. In contrast, high-procrastinators earn 2.9 fewer credits, have a significantly higher dropout rate (5.8 percentage points), and have a worse GPA by the end of the first semester.

TABLE 3—PREDICTIVE POWER OF PROCRASTINATION ON STUDY OUTCOMES

	Dropout	Credits	GPA	Index
	(1)	(2)	(3)	(4)
High procrastinators	0.058*** (0.013)	-2.971*** (0.407)	0.069* (0.029)	-0.232*** (0.041)
Basel. Controls	Yes	Yes	Yes	Yes
Study Program FE	Yes	Yes	Yes	Yes
R-squared	0.04	0.31	0.31	0.08
Observations	2221	2221	1708	2221
<i>Outcome descriptives (low procrastinators)</i>				
mean	0.068	17.2	2.49	0.14
median	0	20	2.50	0.32
s.d.	0.25	10.8	0.70	0.84
min	0	0	1	-3.01
max	1	50	4	0.86

Notes: The table shows the educational outcomes of high procrastinators relative to low procrastinators. Ordinary least squares estimates with robust standard errors in parentheses. High procrastinators are defined by having been assigned an above-median value on the procrastination measure, while low procrastinators have been assigned a below-median value, with the split done within strata. In Germany, the grading system ranges from 1 to 4, with 1 being the highest grade achievable and 4 indicating the lowest passing grade, i.e., the lower the grade, the better. Data set: full population of students of the 2022 cohort. Baseline controls include School GPA, age, time between school graduation and university enrollment, and indicators for being female, having an A-level degree, started the first study attempt, graduated from an out of state school, and graduated abroad. Furthermore, all estimations include study program fixed effects. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C. Take-up of the Learning Journal Interventions

Table 4 summarizes treatment uptake and usage, comparing high-procrastinating students against low-procrastinating students. Across both treatment groups, approximately 50% of students opened the journal link at least once, with roughly 18% of the second treatment group opting to sign the commitment device. On average, it took students around 2 weeks after the start of the treatment to open the link for the first time. Similarly, usage, measured by time spent on the journal screen, was comparable between groups, averaging around 4 minutes¹⁸. We believe that this relatively low usage intensity may stem from students opting for more convenient alternatives to track their study times. The online journaling process may not be as practical as traditional methods like pen and paper or specialized apps. Anecdotal evidence and survey data collected a year after the intervention suggest that treated students, particularly those in the second treatment group, often adopted journaling and goal-setting on paper, finding it faster and more convenient.

Examining uptake and usage through the lens of procrastination tendencies reveals noticeable differences. As anticipated, students exhibiting higher levels of procrastination tendencies demonstrate lower uptake and usage intensity compared to low procrastinators. For instance, while approximately 53.5% of low-procrastinators in both treatment groups accessed the journal link at least once, only 45% of high-procrastinators did so. Similarly, high procrastinators spent less time on the journal online, and it took them, on average, four more days to engage with the treatment.

TABLE 4—TREATMENT TAKE-UP AND USAGE

	Journal Treatment	Journal + Commitment Treatment	Difference (2 vs. 1)	Low Procrastinators	High Procrastinators	Difference (5 vs. 4)
	(1)	(2)	(3)	(4)	(5)	(6)
Open Journal Link	0.493 (0.500)	0.495 (0.500)	0.002 (0.938)	0.535 (0.499)	0.452 (0.498)	-0.083*** (0.001)
Time Elapsed till Take Up (days)	14.811 (22.336) N=365	16.120 (24.241) N=367	1.309 (0.448)	13.552 (20.264)	17.814 (26.402)	4.262** (0.016)
Average Time Spent on Journal (minutes)	3.142 (6.474) N=361	3.843 (7.277) N=364	0.070 (0.171)	4.124 (7.609)	2.727 (5.825)	-1.397** (0.042)
Sign Commitment		0.179 (0.384)		0.200 (0.401)	0.158 (0.366)	-0.042 (0.141)
Observations (total number in each group)	740	741		753	728	

Notes: Columns 1 and 2 present the take-up rate and average usage of the treatment split by treatment groups, while columns 4 and 5 present the take-up rate and average usage of the treatment split by procrastination tendencies of all students assigned to the treatment groups. The reported averages of values for time spent on the journal exclude the top 1% to reduce the influence of extreme outliers. Standard deviation reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

¹⁸ See the appendix for a detailed graphical illustration of average weekly usage split by treatment groups and procrastination tendencies.

III. Treatment Effects

A. Empirical Strategy

This section describes the estimand of the treatment effect and the empirical model that we estimate with our data. Randomization of treatment assignment identifies our empirical model. The treatment indicators $T_{1,i}$ and $T_{2,i}$ are equal to 1 if participation is offered either for the learning journal ($T_{1,i}$) or the learning journal with commitment ($T_{2,i}$) and 0 otherwise. The coefficients α_1 and α_2 in Equation (1) provide the estimates for the ITT effects for both treatment arms. We estimate Equation (1) using a linear ordinary least squares regression and report robust standard errors for the estimates.

$$(1) \quad y_{ij}^k = \alpha_0 + \alpha_1 T_{1,i} + \alpha_2 T_{2,i} + x_i \beta + s_j + \varepsilon_{ij},$$

We use demographic information and pre-treatment outcomes as covariates in our estimations to increase their precision. The vector x_i subsumes pre-specified control variables, which include: a procrastination index, school GPA, age, time between school graduation and university enrolment, and indicators for being female, having an A-level degree, first-time university student, graduated from an out of state school, graduated abroad. Furthermore, a stratified randomization mechanism is employed, we use fixed effects s_j to control for the strata in which each student was randomized into one of the treatment arms, as recommended by Bruhn and McKenzie (2009). The error term is denoted by ε_{ij} .

While the ITT effects capture the impact of offering the journals, including those who never engaged with the treatment, we estimate the TOT to determine the impact among students who actually accessed the learning journal website at least once. For that, we utilize 2SLS to estimate the TOT. Since the control group did not have access to the treatment, the TOT can be calculated as the local average treatment effect by using randomized assignment to the treatment groups as an instrumental variable. We define program take-up as a student who opened the intervention link at least once and thus had engaged with our treatment. The following estimates the TOT:

$$(2) \quad T_{1_takeup_{ij}} = \alpha_{10} + \alpha_{11} T_{1,i} + \alpha_{12} T_{2,i} + x_i \alpha_{12} + s_{1j} + \varepsilon_{1ij}$$

$$(3) \quad T_{2_takeup_{ij}} = \alpha_{10} + \alpha_{12} T_{1,i} + \alpha_{22} T_{2,i} + x_i \alpha_{22} + s_{2j} + \varepsilon_{2ij}$$

$$(4) \quad y_{ij}^k = \gamma_0 + \gamma_1 T_{1_takeup_{ij}} + \gamma_2 T_{2_takeup_{ij}} + x_i \gamma_2 + s_{3j} + \varepsilon_{ij}.$$

The treatment offer was randomized at the individual level, and the invitation letters were sent directly to students and were not discussed within classrooms at the university. Hence, spillover effects on students in the control group are unlikely, supporting the stable unit treatment assumption.

In studying the effects of journaling goals and study time with mental contrasting on education, we expect these effects to vary among students based on their procrastination tendencies. To capture these differences, we conducted a regression analysis focusing on heterogeneous effects for students with high procrastination tendencies. We introduce an interaction term between our treatment and being a procrastinator and estimate the following equation:

$$(5) \quad y_{ij}^k = \alpha_0 + \alpha_1 T_{1,i} + \alpha_2 T_{2,i} + \alpha_3 T_{1,i} * Proc. + \alpha_4 T_{2,i} * Proc. + \alpha_5 Proc. + x_i \beta + s_j + \varepsilon_{ij},$$

where *Proc.* is an indicator that a student is a procrastinator. Specifically, it is equal to 1 if the student has been assigned an above median value on their procrastination measure and 0 otherwise. We again control and include the same specification as in Equation (1).

B. Intention-to-Treat (ITT) Effects

Table 5 presents the ITT effects of assignment to each treatment group on academic outcomes at the end of the first semester. All specifications include the pre-specified baseline controls and strata fixed effects. In the control group, the mean dropout rate is 10 percent, the mean number of credits earned is 15.2 (SD: 11.2), and the mean GPA is 2.57 (SD: 0.69) on the German 1-to-4 scale (where lower values indicate better performance).

We find a clear divergence between the two treatment arms. Assignment to the Journal with Commitment group significantly increases credits earned by 0.85 points (0.075 SD; $p < 0.10$) and improves GPA by 0.086 grade points (0.12 SD; $p < 0.05$). In contrast, assignment to the Journal-Only group produces no statistically significant effects on any outcome, with point estimates that are small and statistically indistinguishable from zero.

The simultaneous improvement in both credits and GPA in the commitment group is noteworthy. A common concern with interventions that increase course-taking is that students may achieve higher credit loads at the expense of lower grades, or conversely, that GPA improvements come from strategic course avoidance. Our results rule out such a quantity-quality trade-off. The combined outcome index (Column 4) is positive for the commitment group (0.065 SD) but does not reach statistical significance. This reflects the index's construction: the standardized inverse-covariance weighting procedure (Schwab et al., 2020) places substantial weight on the dropout outcome, which is unaffected by the treatment. When we instead construct an equal-weighted index, the commitment treatment effect is significant at 0.114 SD ($p < 0.05$).

The null result for the Journal-Only group is instructive. It corroborates a growing body of evidence indicating that planning and scheduling tools, when implemented without binding accountability mechanisms, are often insufficient to alter student behavior (Oreopoulos et al., 2019; Dobronyi et al., 2019; Van Lent, 2019). Our two-arm design allows us to isolate this mechanism directly: while the journal alone replicates the null effects documented in the planning literature, the addition of a commitment device to the identical interface produces significant gains. This sharp contrast implies that the critical ingredient for scalability is not the planning infrastructure itself, but the accountability mechanism it is paired with.

TABLE 5—ITT EFFECTS ON STUDY PERFORMANCE

<i>Dep. Var.:</i>	Dropout	Credits	GPA	Index
	(1)	(2)	(3)	(4)
Journal treatment	0.002 (0.015)	0.265 (0.479)	0.022 (0.035)	-0.002 (0.050)
Journ. & commitment treatment	-0.012 (0.015)	0.848* (0.490)	-0.086** (0.034)	0.065 (0.048)
Basel. Controls	Yes	Yes	Yes	Yes
Strata FE	Yes	Yes	Yes	Yes
R-squared	0.12	0.38	0.38	0.16
Observations	2221	2221	1708	2221
<i>Outcome descriptives (control group)</i>				
mean	0.10	15.2	2.57	0
median	0	16	2.59	0.25
s.d.	0.30	11.2	0.69	1
min	0	0	1	-3.01
max	1	45	4	0.85
<i>(Joint) significant tests (p-values)</i>				
$H_0: J = J\&C = 0$	0.579	0.208	0.003	0.285
$H_0: J = J\&C$	0.340	0.223	0.001	0.170

Notes: The table shows intention-to-treat (ITT) effects on dropout rate, credit points earned, average grade point and an index of the three outcomes at the end of the first academic semester. ‘Journal treatment’ denotes random assignment to the first treatment group. ‘Journ. & commitment treatment’ denotes random assignment to the second treatment group. Ordinary least squares estimates with robust standard errors in parentheses. In Germany, the grading system ranges from 1 to 4, with 1 being the highest grade achievable and 4 indicating the lowest passing grade, i.e., the lower the grade the better. Data set: full population of students of the 2022 cohort. Baseline controls include Procrastination Index, School GPA, age, time between school graduation and university enrollment, and indicators for being female, having an A-level degree, started the first study attempt, graduated from an out of state school, graduated abroad. Furthermore, all estimations include strata fixed effects. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C. TOT Effects

The ITT estimates reflect the effect of offering the intervention, including students who never engaged with it. To estimate the effect of engaging with the treatment, we employ two-stage least squares, using random assignment as an instrument for treatment take-up, as specified in Equations (2)–(4). We define a student as treated if they opened the intervention link at least once during the semester. This is a minimal definition of engagement, as it captures any exposure to the journal, regardless of subsequent usage intensity.

First-stage results (Table 6, Panel B) confirm the strength of our instrument (F-statistic exceeds 500). Compliance rates are nearly identical across treatment arms: 49.3 percent in the journal group and 49.6 percent in the commitment group. Crucially, the symmetry in compliance rates implies that the differential effects observed in the second stage are driven by the content of the commitment device (treatment mechanism) rather than by differential selection into usage. Analysis of complier characteristics (Table A1) indicates some positive selection, as compliers exhibit slightly lower

baseline procrastination and higher high school GPAs, though socio-economic backgrounds are comparable.

Table 6 presents the TOT effects. Treated students in the commitment arm earn 1.71 more credits (0.15 SD; $p = 0.07$) and achieve a GPA improvement of 0.157 grade points (0.22 SD; $p < 0.01$). The journal-only group again shows no significant effects. By the benchmarks proposed by Kraft (2020), both the credit and GPA estimates for the commitment group fall within the range of medium-sized educational interventions.

TABLE 6—TOT EFFECTS ON STUDY PERFORMANCE

<i>Dep. Var.:</i>	Dropout	Credits	GPA	Index
	(1)	(2)	(3)	(4)
Journal treatment	0.004 (0.030)	0.541 (0.934)	0.042 (0.063)	-0.004 (0.098)
Journ. & commitment treatment	-0.024 (0.029)	1.707* (0.945)	-0.157*** (0.059)	0.131 (0.093)
Basel. Controls	Yes	Yes	Yes	Yes
Strata FE	Yes	Yes	Yes	Yes
R-squared	0.12	0.39	0.38	0.16
Observations	2221	2221	1708	2221
<i>Outcome descriptives (control group)</i>				
mean	0.10	15.2	2.57	0
median	0	16	2.59	0.25
s.d.	0.30	11.2	0.69	1
min	0	0	1	-3.01
max	1	45	4	0.85
<i>First Stage Dep. Var.: Open Treatment Link ^(B)</i>				
Journal treatment	0.493*** (0.017)	0.493*** (0.017)	0.493*** (0.017)	0.493*** (0.017)
Journ. & commitment treatment	0.496*** (0.018)	0.496*** (0.018)	0.496*** (0.018)	0.496*** (0.018)
1 st Stage F-Statistic	669.6	669.6	587	669.6

Notes: The table shows the treatment on the treated effects (TOT) effects on dropout rate, credit points earned, average grade point, and an index of the three outcomes at the end of the first academic semester. IV estimates using 2SLS are presented. The endogenous variable is an indicator variable for whether a student opened the intervention link. The instruments are indicator variables for the random assignment to the treatment groups. 'Journal treatment' denotes random assignment to the first treatment group. 'Journ. & commitment treatment' denotes random assignment to the second treatment group. All specifications are identical to Table 5. Panel B at the bottom of the table presents the first stage estimates and first stage F-Statistic. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

To put these magnitudes in practical terms: Given that a standard course at the university is worth 5 ECTS, the TOT effect on credits implies that roughly one in three treated students completed an additional course. The GPA improvement of 0.157 points represents approximately one-quarter of

a within-group standard deviation, a gain that, if sustained, could meaningfully affect students' academic trajectories (Attewell et al., 2012) and labor market prospects (Jones and Jackson, 1990). These estimates align with recent findings on commitment devices in education: Himmeler et al. (2019) report treatment effects of roughly 0.25 SD on academic progress for students who voluntarily commit to study plans, while Patterson (2018) documents GPA improvements of 0.29 SD from a commitment device designed to limit internet distractions.

D. Heterogeneous Treatment Effect Based on Procrastinating Behavior

Our intervention targets several dimensions of self-regulation—goal setting, mental contrasting, reminders, and commitment—that are central to the behavioral determinants of procrastination. A natural question is whether the treatment effects documented above vary with students' baseline procrastination tendencies. On the one hand, students who procrastinate more may stand to gain the most from a structured commitment mechanism that directly counteracts present bias and poor self-regulation. On the other hand, students with lower procrastination tendencies may also benefit, as the planning and reflective components of the journal can enhance already functional self-regulatory skills. To examine this heterogeneity, we interact the treatment indicators with an above-median indicator of baseline procrastination, as specified in Equation (5).

Table 7 presents the results. All specifications include baseline controls and strata fixed effects. Consistent with expectations, high-procrastinators perform significantly worse than low-procrastinators across all achievement outcomes. Turning to the treatment interactions, we find a marginally significant interaction between the commitment treatment and procrastination status for GPA (0.117; $p < 0.10$), indicating that the GPA benefit of the commitment treatment is concentrated among low-procrastinators. The interaction term for credits earned is positive but not statistically significant (0.670; $p > 0.10$), suggesting that while the pattern of differential effects on credit accumulation is directionally consistent, the data do not allow us to reject equal treatment effects across subgroups for this outcome.

To characterize the treatment effects within each subgroup, we report marginal effects at the bottom of Table 7. Among high-procrastinators, students assigned to the journal and commitment group earn 1.18 more credit points than their counterparts in the control group ($p < 0.10$). Among low-procrastinators, the commitment treatment yields a significant improvement in GPA of 0.138 grade points ($p < 0.01$). Notably, we also find that low-procrastinators in the journal-only group earn 1.10 more credits ($p < 0.10$), an effect that is absent for them in the commitment group. While the specific reduction in dropout rates for this subgroup is not statistically significant in isolation, the combined outcome index is significantly higher for low-procrastinators in the commitment group (0.123 SD; $p < 0.05$).

These findings suggest that the journal and commitment intervention operates through distinct channels depending on students' baseline self-regulation. Among high-procrastinators, the primary margin of improvement is credit accumulation rather than GPA. Evidence from Table 9 sheds light

on the underlying mechanism: high-procrastinators in the commitment group sign up for significantly more credits early in the semester (1.7 additional credits; $p < 0.05$), a task that requires action during the course registration period. Combined with the significant reduction in self-reported procrastination tendencies among this group (Table 9, column 1), these results indicate that the commitment device facilitates short-term task initiation. Thereby, prompting procrastinators to register for courses they might otherwise delay, which in turn translates into higher credit attainment by semester's end.

Among low-procrastinators, the treatment instead improves the quality of academic output, as reflected in GPA gains. Several features of the data help explain this pattern. First, low-procrastinators exhibit higher take-up of the commitment device (20% vs. 15.8%), suggesting they utilize the tool to support sustained self-regulation rather than simply to overcome initial inertia. Second, because low-procrastinators already accumulate credits at higher rates, there is limited scope for further gains along this margin. Third, the GPA improvements among this subgroup are consistent with the sustained use of the journal's reflective and planning components over the course of the semester, which can enhance effective learning time and study quality even among students who do not face severe self-regulation deficits. The mental contrasting component, in particular, may promote deeper self-reflection on study strategies, improving the learning process in ways that manifest as higher grades rather than additional course completions.

Taken together, the heterogeneity analysis reveals that the intervention's commitment mechanism serves a dual function: it helps high-procrastinators overcome delays in task initiation, and it supports low-procrastinators in maintaining and improving the quality of their academic engagement.

TABLE 7—ITT EFFECTS ON STUDY PERFORMANCE BY PROCRASTINATION

<i>Dep. Var.:</i>	Dropout	Credits	GPA	Index
	(1)	(2)	(3)	(4)
Journal treatment	-0.020 (0.019)	1.102* (0.661)	0.051 (0.048)	0.064 (0.063)
Journ. & commitment treatment	-0.029 (0.018)	0.511 (0.678)	-0.138*** (0.046)	0.123** (0.060)
1[Above median procrastination]	0.246*** (0.121)	-12.692*** (3.178)	-0.291 (0.405)	-0.846** (0.385)
J Treatm. x 1[Procr. > median]	0.043 (0.031)	-1.676* (0.963)	0.067 (0.071)	-0.130 (0.102)
J&C Treatm. x 1[Procr. > median]	0.034 (0.030)	0.670 (0.987)	0.117* (0.068)	-0.120 (0.098)
Basel. Controls	Yes	Yes	Yes	Yes
Strata FE	Yes	Yes	Yes	Yes
R-squared	0.11	0.38	0.38	0.15
Observations	2221	2221	1708	2221
<i>Journal treatment effect for:</i>				
low procrastination students	-0.020 (0.019)	1.102* (0.662)	0.051 (0.048)	0.064 (0.063)
high procrastination students	0.024 (0.025)	-0.573 (0.694)	-0.016 (0.052)	-0.066 (0.080)
<i>Journ. & commitment treatment effect for:</i>				
low procrastination students	-0.029 (0.018)	0.511 (0.678)	-0.138*** (0.046)	0.123** (0.060)
high procrastination students	0.005 (0.024)	1.181* (0.716)	0.021 (0.050)	0.003 (0.077)

Notes: The table shows intention-to-treat (ITT) effects on dropout rate, credit points earned, average grade point and an index of the three outcomes at the end of the first academic semester. All estimates include an interaction between treatment assignment and whether a student is a procrastinator. 'Journal treatment' denotes random assignment to the first treatment group. 'Journ. & commitment treatment' denotes random assignment to the second treatment group. Ordinary least squares estimates with robust standard errors in parentheses. High procrastinators are defined by having been assigned an above-median value on the procrastination measure, while low procrastinators have been assigned a below-median value, with the split done within strata. In Germany, the grading system ranges from 1 to 4, with 1 being the highest grade achievable and 4 indicating the lowest passing grade, i.e., the lower the grade the better. Data set: full population of students of the 2022 cohort. Baseline controls include Procrastination Index, School GPA, age, time between school graduation and university enrollment, and indicators for being female, having an A-level degree, started the first study attempt, graduated from an out of state school, graduated abroad. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

E. Analysis of Mechanisms

This section examines the mechanisms underlying the observed improvements in academic achievement in the commitment treatment group. We draw on survey data and learning journal inputs to assess whether changes in procrastination tendencies and study behavior can account for the effects documented above. Since these outcomes rely partly on self-reported data from voluntary surveys, we first address potential selection concerns before turning to the results.

Survey balance and selection.—Table 8 reports baseline characteristics of survey respondents by experimental group. Observable characteristics are balanced across arms, and we cannot reject

equality across background variables. However, response rates differ substantially: the control group exhibits a higher response rate (approximately 29 percent) than the treatment groups (13–16 percent). A plausible explanation is that treated students may have mistaken survey invitations for routine journal reminders, leading to lower participation.

Differential nonresponse raises the possibility of selection on unobservables, such as motivation. To mitigate this concern, the mechanism analysis focuses primarily on comparisons between the two treatment groups, which display nearly identical response rates and balanced observable characteristics. This comparison isolates the marginal contribution of the commitment device while reducing concerns about differential survey selection relative to the control group. Nevertheless, because these measures are observed for a selected subsample, we interpret the results as supportive rather than definitive evidence on mechanisms.

TABLE 8—SURVEY RESPONDER CHARACTERISTICS & BALANCING

	Control	Journal Treatment	Journal + Commitment Treatment	Joint Orthogonality Test (p-value)
<i>Socio-economic background</i>				
1[Female]	0.44	0.41	0.47	0.678
Age at enrollment (years)	21.31	20.77	20.96	0.224
1[A-Level]	0.54	0.52	0.67	0.094
1[schooling outside of Bavaria]	0.13	0.06	0.12	0.148
1[foreign schooling]	0.02	0.03	0.05	0.495
<i>Study characteristics</i>				
Procrastination index (std.)	-0.184	-0.117	-0.123	0.793
Matriculation date (average)	Aug 09, 2022	Aug 10, 2022	Aug 07, 2022	0.444
Application date (average)	Jun 18, 2022	Jun 17, 2022	Jun 21, 2022	0.665
1[Missing application date imputed]	0.01	0.06	0.04	0.088
High school GPA (German scale: 1 very good - 4 very poor)	2.29	2.26	2.25	0.849
1[Missing high school GPA imputed]	0.02	0.01	0.03	0.795
1[first study program]	0.82	0.79	0.82	0.780
Time between school degree and enrollment (years)	1.55	1.43	1.30	0.500
Observation per group	215	116	97	

Notes: The table shows the survey responders' means of main outcomes at baseline for the control group and two treatment groups. The final column reports the p-value of an F-test on the treatment arms not predicting the respective baseline outcome or any of the further variables. Variables in brackets with a leading digit 1 (1[...]) are binary variables.

Table 9 presents intention-to-treat estimates comparing the Journal + Commitment group to the Journal-only group. The timing of the semester is central to interpreting the mechanism. The first survey is administered during the exam registration period in early November, the same window in which students must sign up for courses. This timeline allows us to test whether reductions in procrastination translate into task initiation at the moment when key academic decisions are made.

Assignment to the commitment treatment reduces self-reported procrastination by 0.93 points in the first survey, corresponding to a 0.33 standard deviation reduction ($p < 0.05$). Panel B shows that this effect is driven entirely by high-procrastinators. For this group, procrastination declines by 1.21 points (0.43 SD; $p < 0.10$) in the first survey and by 1.71 points (0.57 SD; $p < 0.10$) in the second survey. Low-procrastinators exhibit no statistically significant change. The concentration of effects

among high-procrastinators is consistent with the commitment device increasing the psychological cost of delay for those most prone to present-biased inaction.

Crucially, the procrastination reduction coincides with a change in observable behavior. Credit sign-up during the registration window represents a discrete, deadline-driven task that requires early-semester action. High-procrastinators assigned to the commitment treatment sign up for 1.75 additional credit points relative to their counterparts in the journal-only group (0.16 SD; $p < 0.05$). Low-procrastinators show no change. The temporal and subgroup alignment is clear: procrastination falls among high-procrastinators during the registration period, and credit sign-up rises for the same group in the same window. This pattern provides coherent evidence that the commitment device counteracts present-biased delay at a critical decision point, prompting earlier task initiation. Consistent with this interpretation, high-procrastinators in the commitment group earn more credits by the end of the semester, reflecting the downstream consequences of higher initial enrollment (Behlen, Himmler & Jäckle, 2023).

We find no evidence that the commitment device increases total study time; weekly study hours remain statistically indistinguishable across treatment arms. This rules out the possibility that students simply worked longer. Rather, the evidence suggests improved quality of learning. By reducing the behavioral inefficiencies associated with procrastination, the intervention appears to have enabled students to learn more effectively within the same time constraints.

For low-procrastinators, the mechanism operates differently. This group shows neither a significant reduction in procrastination nor a change in credit sign-up, consistent with already low baseline procrastination and high enrollment rates. Nevertheless, as shown in the heterogeneous effects analysis above, low-procrastinators in the commitment group achieve significantly higher GPAs. A plausible explanation lies in differential engagement with the commitment device: low-procrastinators are more likely to sign the voluntary target agreement (20 percent versus 15.8 percent; Table 4), suggesting that they actively use the commitment mechanism to support sustained self-regulation. In combination with the journal's reflective and planning components, this higher take-up may reinforce consistent study habits throughout the semester, leading to improvements in the quality of academic performance rather than increases in course enrollment.

Taken together, the evidence supports a dual mechanism. The commitment device addresses distinct binding constraints for different students. For high-procrastinators, it solves a problem of initiation (reducing procrastination by 0.43–0.57 SD), which boosts credit accumulation. For low-procrastinators, it supports a process of optimization, reinforcing habits that improve GPA.

TABLE 9—ITT EFFECTS ON PROCRASTINATION PROXY, CP SIGN-UP, AND STUDY TIME

<i>Dep. Var.:</i>	Procrastination S1	Procrastination S2	CP Sign-up	Study time
	(1)	(2)	(3)	(4)
Journ. & commitment treatment	-0.926** (0.452)	-0.484 (0.718)	0.780 (0.494)	2.246 (1.726)
Basel. Controls	Yes	Yes	Yes	Yes
Study Program FE	Yes	Yes	Yes	Yes
R-squared	0.30	0.42	0.25	0.11
Observations	179	92	1477	322
Mean in journal treatment group	5.85 (<i>sd.:2.8</i>)	5.90 (<i>sd.:3.0</i>)	24.5 (<i>sd.:11.2</i>)	24.6 (<i>sd.:12.5</i>)
<i>Marginal Effects by Baseline Procrastination Tendencies^(B)</i>				
T2: Low-procrastinator	-0.513 (0.608)	0.675 (0.994)	-0.202 (0.623)	1.350 (1.990)
T2: High-procrastinator	-1.214* (0.626)	-1.707* (0.936)	1.752** (0.800)	3.081 (2.708)

Notes: 'Journ. & commitment treatment' denotes random assignment to the second treatment group. The comparison group is the first treatment group. Column 1 and 2 show intention-to-treat (ITT) effects of being assigned to the journal and commitment group on self-reported post-intervention procrastination tendencies, in terms of postponing learning activities measured on an 11-point Likert scale, from the first and second surveys, respectively. "CP Sign-up" refers to the number of credit points signed up for by students approximately a month after the beginning of the semester. Column 4 shows the average self-reported study time per week as pooled from the learning journal and 2 surveys. Panel B at the bottom half of the table reports the respective marginal effects based on baseline procrastination tendencies. Ordinary least squares estimates with robust standard errors in parentheses. We use study program fixed effects instead of strata fixed effects to avoid inflated measurements from small strata sample sizes. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

F. Side Effects of the Intervention

Possible side effects of the treatment might positively or negatively affect student well-being. On the one hand, the treatment could decrease well-being by increasing perceived pressure to perform. On the other hand, it might alleviate stress by reducing procrastination, thereby improving well-being. We used two post-intervention surveys to investigate any treatment side effects on various well-being outcomes and reported the pooled results in Table 10. Overall, we find no significant differences in well-being indicators like life satisfaction, pressure, and stress. The absence of adverse well-being effects suggests that the academic gains documented above are not achieved at the cost of increased student stress.

TABLE 10—EFFECTS ON WELL-BEING OUTCOMES

	Satisfaction	Pressure	Stress	Well-being index
	(1)	(2)	(3)	(4)
Journal treatment	-0.060 (0.109)	-0.279 (0.185)	-0.163 (0.174)	-0.063 (0.119)
Journ. & commitment treatment	0.146 (0.102)	0.134 (0.197)	-0.248 (0.189)	0.126 (0.119)
Basel. Controls	Yes	Yes	Yes	Yes
Study program FE	Yes	Yes	Yes	Yes
r ²	0.18	0.16	0.15	0.05
Observations	425	407	409	426
<i>Outcome descriptive (control group)</i>				
mean	-0.024	4.21	4.66	0
median	0.071	4	5	0.072
s.d.	0.94	1.51	1.40	1
min	-3.39	1	1	-4.56
max	1.98	7	7	2.34
<i>(Joint) significant tests (p-values)</i>				
H ₀ : J = J&C = 0	0.145	0.149	0.367	0.319
H ₀ : J = J&C	0.062	0.063	0.687	0.139

Notes: The table shows intention-to-treat (ITT) effects on well-being outcomes. ‘Journal treatment’ denotes random assignment to the first treatment group. ‘Journ. & commitment treatment’ denotes random assignment to the second treatment group. Ordinary least squares estimates with robust standard errors in parentheses. All dependent variables are constructed based on a combination of student responses in 2 surveys throughout the semester. Baseline controls include Procrastination Index, School GPA, age, time between school graduation and university enrollment, and indicators for being female, having an A-level degree, started the first study attempt, graduated from an out of state school, graduated abroad. All estimations include study program fixed effects. * p < 0.10, ** p < 0.05, *** p < 0.01.

IV. Conclusion

This paper provides experimental evidence that a low-cost, scalable behavioral intervention—a structured learning journal combined with a soft commitment device—can improve academic performance in higher education. In a pre-registered field experiment, we show that offering the journal together with a voluntary commitment mechanism increases earned credits by 0.85 points (0.075 SD) and improves GPA by 0.12 SD in the first semester. The journal alone has no statistically significant effects. These gains are achieved at minimal cost, requiring only a simple online platform and automated reminders, and generate no detectable adverse effects on student well-being.

Two central insights emerge. First, planning tools appear ineffective in isolation but become effective when paired with an accountability mechanism. The null effects of the journal-only arm, contrasted with the significant gains in the journal-plus-commitment arm, provide within-study evidence that structured goal setting requires an embedded commitment component to translate intentions into action. Importantly, the commitment device is soft: it involves no financial stakes, monitoring, or enforcement. The psychological cost of deviating from a self-set agreement is sufficient to generate measurable improvements in academic outcomes. This finding helps reconcile mixed results in the literature on planning interventions and highlights a scalable pathway for enhancing their effectiveness.

Second, the intervention operates through distinct behavioral channels across students. Among high-procrastinators, the commitment device reduces procrastination during the early-semester registration period and increases credit sign-up at that critical decision point, leading to higher credit accumulation by semester's end. The effect is concentrated in the timing of action rather than the total quantity of study time. Among low-procrastinators, credit sign-up and reported procrastination remain largely unchanged, yet GPA improves, consistent with reinforced self-regulation and improvements in the quality or consistency of effort over the semester. The same intervention thus addresses different behavioral bottlenecks, overcoming inertia for those prone to delay and strengthening discipline for those already relatively self-regulated.

We note several limitations that suggest avenues for future research. First, our primary data covers only the first semester. Thus, whether repeated exposure to commitment devices across multiple semesters can produce durable habit formation, or if effects fade as the novelty wears off, remains an open question. Second, engagement poses a challenge. Our study is not able to track whether students continued journaling outside the online portal, making it challenging to assess engagement levels accurately. Approximately half of the assigned students accessed the digital tool, and anecdotal evidence suggests that some adopted journaling and goal-setting practices using pen and paper, which our administrative data cannot capture. Future implementations could explore alternative, user-friendly delivery formats, such as physical journals or integration with university learning management systems, to reduce friction and capture this offline behavior.

Despite these limitations, the results highlight a highly favorable benefit-to-cost ratio for policy implementation. The TOT estimates indicate that for students who actively engage, the impact is substantial, equivalent to one in three students completing an additional course. Given the high returns to degree completion and the low baseline graduation rates in many educational systems, widely deploying such commitment-enhanced tools could yield significant efficiency gains. Ultimately, our findings suggest that the efficacy of digital interventions hinges not on the complexity of the planning tool, but on the presence of psychological accountability.

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Appendix

A. Descriptive Analysis of Journal Usage

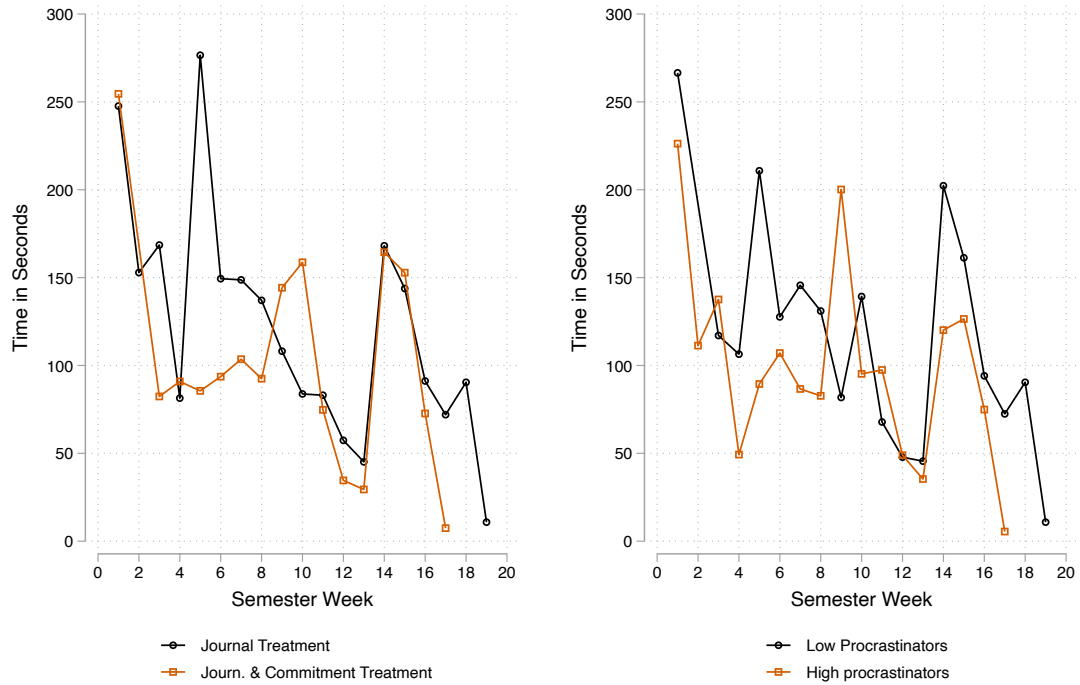


FIGURE A1. AVERAGE WEEKLY USAGE OF LEARNING JOURNAL

Notes: The Figure shows the average screen time recorded per week in seconds. The left-side figure compares average usage of the two treatment groups. The figure on the right-side presents weekly usage of the learning journal separated by procrastination tendencies. Weeks 12 and 13 are the end of the year holidays. The graphs were restricted to the 95th percentile to limit the effect of outliers on the distribution. Outliers in the time spent on the journal could occur if a student forgets to close the journal webpage for prolonged time.

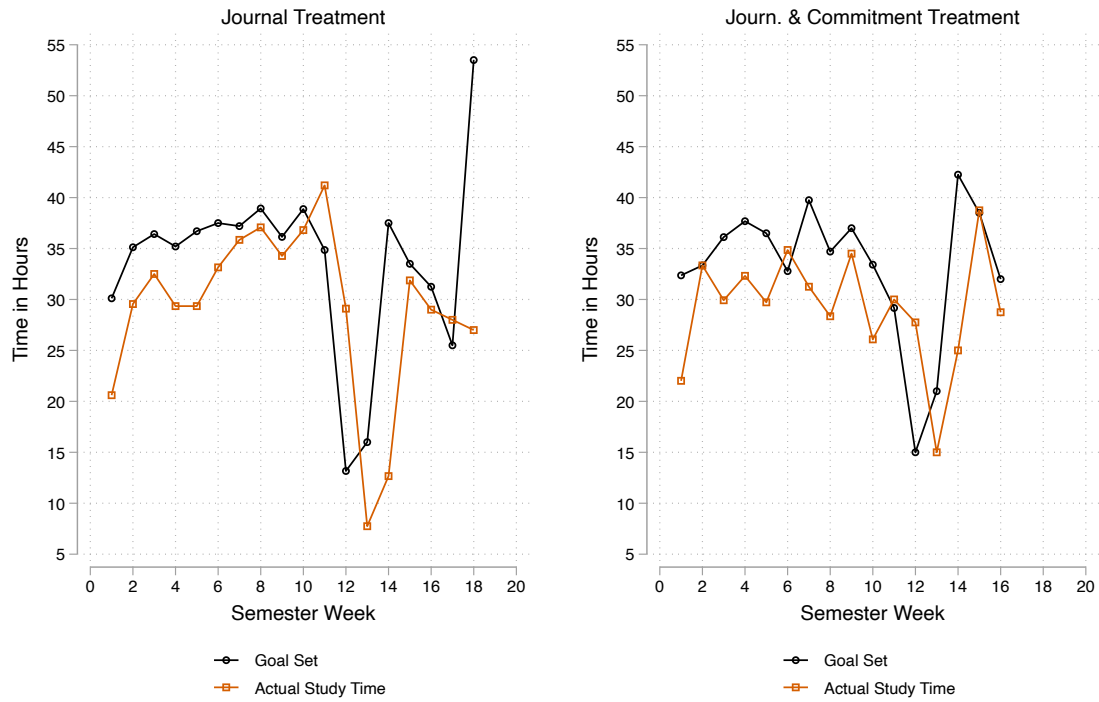


FIGURE A2. AVERAGE WEEKLY INPUTS IN THE LEARNING JOURNAL BY TREATMENT GROUP

Notes: The figure shows the average goal set and actual study time per week in hours as imputed in the learning journal. The left-side figure presents the weekly average goals and study time in the first treatment group. The figure on the right side presents the weekly average goals and study time in the second treatment group. Weeks 12 and 13 are the end-of-year holidays.

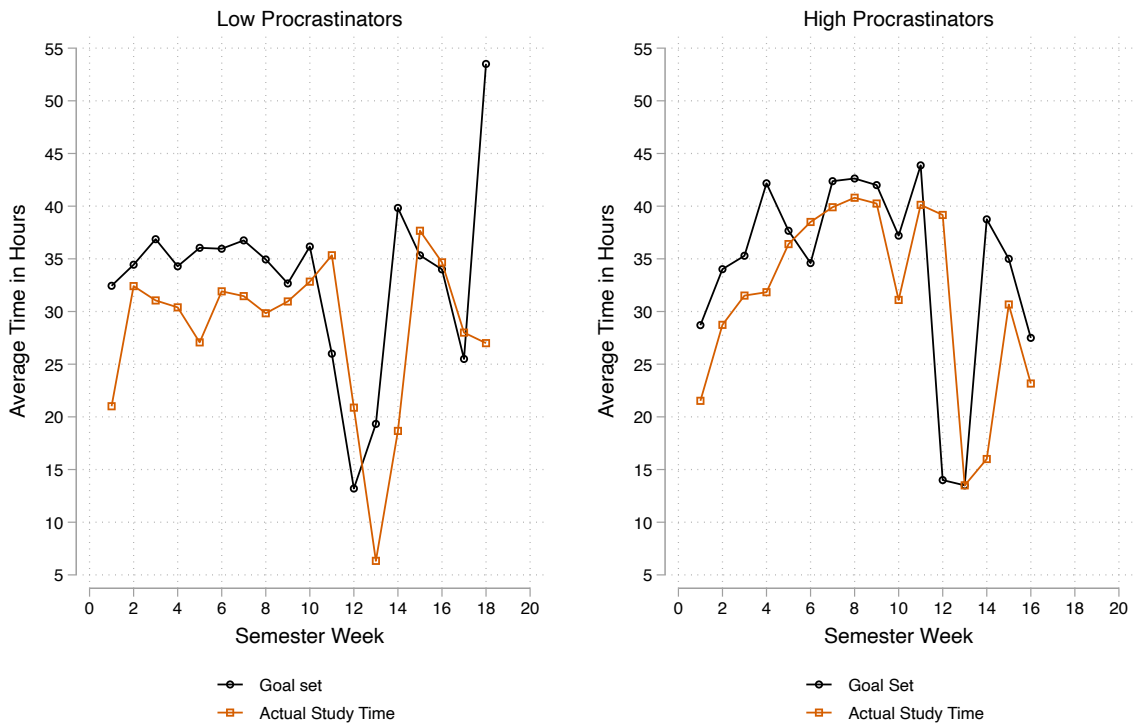


FIGURE A3. AVERAGE WEEKLY INPUTS IN THE LEARNING JOURNAL BY PROCRASTINATION TENDENCIES

Notes: The figure shows the average goal set and actual study time per week in hours as imputed in the learning journal. The left-side figure presents the weekly average goals and study time for low procrastinators. The figure on the right side presents the weekly average goals and study time for high procrastinators. Weeks 12 and 13 are the end-of-year holidays.

B. TOT EFFECTS

TABLE A1—

BALANCING TABLE FOR COMPLIERS AND NON-COMPLIERS (T1 & T2 COMPLIERS= OPEN LINK)

	Non-Compliers	Compliers Journal Treatment	Compliers Commitment Treatment	Joint Orthogonality Test (p-value)	N
<i>Socio-economic background</i>					
1[Female]	0.35	0.41	0.38	0.152	2221
Age at enrollment (years)	21.44	21.10	21.23	0.116	2221
1[A-Level]	0.51	0.52	0.55	0.336	2221
1[schooling outside of Bavaria]	0.09	0.10	0.09	0.856	2221
1[foreign schooling]	0.03	0.04	0.04	0.414	2221
<i>Study characteristics</i>					
Procrastination index (std.)	0.070	-0.178	-0.110	0.000	2221
Matriculation date (average)	Aug 13, 2022	Aug 08, 2022	Aug 09, 2022	0.000	2221
Application date (average)	Jun 24, 2022	Jun 21, 2022	Jun 21, 2022	0.002	2216
1[Missing application date imputed]	0.07	0.04	0.04	0.021	2221
High school GPA (German scale: 1 very good - 4 very poor)	2.49	2.44	2.41	0.041	2221
1[Missing high school GPA imputed]	0.01	0.01	0.02	0.643	2221
1[first study program]	0.74	0.808	0.763	0.021	2221
Time between school degree and enrollment (years)	1.724	1.341	1.492	0.006	2221
Observations per group	1489	365	367		

Notes: The table shows the compliers and non-compliers means of the main outcomes at baseline. Compliance is defined as open journal link for both treatment groups. Variables in brackets with a leading digit 1 (1[...]) are binary variables. Last row reports the number of students in each group. Small differences in group sizes arise because randomization was conducted within strata blocks, with some strata blocks having block sizes that were not multiples of three (the number of randomization groups).

C. Experimental Design

The learning journal is offered to students of both treatment groups (T1 and T2) and consists of the following elements. Starting from the second week of the semester until the end of the exam period of the semester, the learning journal gives students the opportunity to reflect on their learning behavior of the last week and to plan ahead until the end of the next week. Specifically, students can enter the following type of information in each week:

1. Study times on each day of the last week in hours, separately for attendance and self-studying. From the second week onward students receive a notification that compares their reported study time to the goals that they initially set for the week (cf. next point).
2. Study time goals for each day of the upcoming week in hours, separately for attendance and self-studying.
3. Positive results they expect from achieving their study time goals.
4. What prevented them from achieving their goals in the last week or could prevent them from achieving their goals in the upcoming week.
5. “if-then” statements on how they will deal with the obstacles that they listed under 4 in the upcoming week.

In each week of the semester, students of the two treatment groups will receive an email reminding them to use the learning journal.

For students in treatment group T2, right at the beginning of the learning journal, we additionally provide the offer to sign a target agreement, according to which students declare “that I am committed to the study times formulated in the learning journal, and in order to study successfully with adhere to these study times”, i.e., the study time goals that they set under question 2. Students are informed that signing the agreement is voluntary and that failing to meet the goals has no consequences under study or examination regulations. The text, however, states that by signing the agreement they show that they are committed to their self-set goals and that this will increase the likelihood of achieving them. Following the taxonomy of Bryan et al. (2010), the target agreement thus constitutes a soft commitment device. Students can sign the agreement digitally by ticking a checkbox and entering their first and last name next to it. In case students have signed the agreement, in each week, question 2 includes a notification that reminds them about their commitment.

D. Randomization Procedure

Students were assigned to one of the three experimental groups within strata that we constructed based on the following steps:

1. We split the sample based on the students’ study programs resulting in 21 strata (cf. Table 1).
2. To account for differences in students’ ability, within each study program, we split our sample at the median of the high school GPA distribution.¹⁹ If this split results in two groups that include at least 24 students, we keep the two groups (14 study programs). For the other programs, we proceed without splitting the sample by high school GPA (7 study programs).²⁰ The second step results in 35 strata.
3. The goal of our final step is to achieve balance along a procrastination proxy that we construct based on the date of application to the university and the date of enrollment at the university, which is our main heterogeneity dimension of interest.²¹ To do so, within each of the 35 strata that resulted after the first two steps, we order students according to their value of the procrastination index (ascending order) and split the sample into an even number of smaller groups of at least 12 students.²² “Surplus” students are distributed across the two strata in the

¹⁹ The high school GPA was missing for 42 observations. To keep the sample complete, we imputed those values based on a linear regression of the high school GPA on age, a female dummy, time since high school graduation in years, a high school degree Abitur dummy, place of high school degree dummies (foreign and other federal state, the reference group federal state is Bavaria), the procrastination index (cf. Footnote 7), a first semester at any university dummy as well as study program dummies, and the interaction of the study program dummies with the other variables.

²⁰ In three study programs (Applied Materials Science, Civil Engineering, Business Administration), we made a small change to the median split to achieve a more equal distribution of the two groups. E.g., in Business Administration the median split resulted in groups of 221 and 170 students, and we changed the cutoff such that the final split was 194 and 197 students.

²¹ To construct the proxy, we used *Stata*’s *swindex* command by Schwab et al. (2020) to calculate the standardized inverse-covariance weighted average (Anderson, 2008) of the date of application for the study program and the date of enrollment. The date of enrollment was first standardized within study programs due to differences in the timelines of the enrollment periods between study programs. For the application date, we first imputed 130 missing observations based on a linear regression of the application date on the application number, study program dummies, and the interaction of the study program dummies with the application number (5 remaining missings are accounted for by the *swindex* command). Afterwards, we standardized the application date within study programs and information on whether students graduated from high school before or after the beginning of the application period, since the application behavior of those two groups differs from each other.

²² In Energy Process Engineering, which only has 23 students, we split the sample into one group of 12 and one group of 11 students. Due to the removal of duplicates by the university (see Section 2), in the final sample, Social Work also has two strata with only 11 students.

middle of the distribution starting with the stratum with the lower procrastination tendency, e.g., if there are 50 students, we create 4 strata with 12, 13, 13, and 12 students. However, if there are exactly 3 surplus students, they are allocated to one stratum together. E.g., if there are 51 students, we create 4 strata with 12, 15, 12, and 12 students. We create an even number of strata in each study program to be able to split students/strata in each study program into a group of students with low procrastination tendencies (N=1,130) and one with high procrastination tendencies (N=1,091). In total, this process results in 156 Strata.

E. Outcome Variable Definition

Variable	Definition
Passed Course Credits	Credit points obtained in the first semester net of credits granted for an internship and transferred credits.
GPA	Grade point average at the end of the respective semester (1=best, 4=worst); failed exams are not included in calculation
Dropout	Indicator for having dropped out of the study program before or in the respective semester.
Academic Index	Index using a standardized inverse-covariance weighted average (includes: credits earned, GPA, and Dropout)
Procrastination	Measures procrastination based on using a standardized inverse-covariance weighted average
Credit Signed Up	Credit points signed up for in the first semester
Credit Attempted	Credit points attempted in the first semester
Hours Invested in Studying	Number of hours spent studying during a week
Well-being Outcome	Index using a standardized inverse-covariance weighted average of responses from 5 questions on well-being (see survey questions for more information)
<i>Treatment Variables:</i>	
Journal Treatment	Random assignment to the first treatment group
Journal + Commitment	Random assignment to the second treatment group

Studieren mit Plan

Guten Tag **Vorname Nachname**,

herzlich willkommen an unserer Hochschule! Wir freuen uns, Sie bei uns begrüßen zu dürfen. Unser Programm **Studieren mit Plan** möchte Sie während Ihres ersten Semesters an der TH Nürnberg unterstützen. Hierzu testen wir aktuell verschiedene Angebote:¹

Studieren mit Plan für Vorname Nachname, Matrikelnummer 1234567:

Ihr Studiengang **studiengang** ist ein Präsenzstudiengang, der in Vollzeit angeboten wird. Das heißt, Ihr Studium ist so konzipiert, dass sich für einen Abschluss in der Regelstudienzeit durch alle studienbezogenen Tätigkeiten ca. eine 40-Stundenwoche ergibt. Dieser Zeitumfang ist ein Anhaltspunkt, der für typische Studierende gilt und in Abhängigkeit von Vorwissen, Talenten und den persönlichen Ansprüchen und Lebensumständen unter- oder überschritten werden kann.

Natürlich wollen Sie Ihre Zeit optimal einsetzen. Nutzen Sie dazu unser **Lern-Journal!**

Das Journal begleitet Sie durch das erste Semester. Sie reflektieren darin kurz Ihr Studier- und Lernverhalten und planen voraus.

Warum ein Journal führen? Wissenschaftliche Erkenntnisse zeigen, dass wir unsere Ziele eher erreichen, wenn wir sie aufschreiben und unseren Fortschritt dokumentieren.

Erhöhen Sie Ihre Wahrscheinlichkeit, erfolgreich zu studieren, indem Sie dem QR-Code oder dem Link folgen: [Beispiellink](#)



Studieren mit Plan ist keine Einbahnstraße: geben Sie uns **Feedback in zwei 3-Minuten Surveys!** Während des Semesters werden wir uns zwei Mal mit Fragen zum Studium melden. Wir freuen uns über Ihre Rückmeldung.

Bitte beachten Sie auch unsere Hinweise und Beratungsangebote auf der nächsten Seite.

Wir wünschen Ihnen viel Freude und Erfolg beim Studieren an unserer Hochschule!

Mit freundlichen Grüßen

Ihr KoSIMA-Team

FIGURE A4. TREATMENT INTERVENTION LETTER

Notes: The figure shows a screenshot of an example of the treatment letter sent to all students in the treatment groups.

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Studieren mit Plan für Vorname Nachname, Matrikelnummer 1234567:

Ihr Studiengang **studiengang** ist ein Präsenzstudiengang, der in Vollzeit angeboten wird. Das heißt, Ihr Studium ist so konzipiert, dass sich für einen Abschluss in der Regelstudienzeit durch alle studienbezogenen Tätigkeiten ca. eine 40-Stundenwoche ergibt. Dieser Zeitumfang ist ein Anhaltspunkt, der für typische Studierende gilt und in Abhängigkeit von Vorwissen, Talenten und den persönlichen Ansprüchen und Lebensumständen unter- oder überschritten werden kann.

Studieren mit Plan ist keine Einbahnstraße: geben Sie uns in **zwei 3-Minuten Surveys Feedback!** Während des Semesters werden wir uns zwei Mal mit Fragen zum Studium melden. Wir freuen uns über Ihre Rückmeldung.

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Wir wünschen Ihnen viel Freude und Erfolg beim Studieren an unserer Hochschule!

Mit freundlichen Grüßen

Ihr KoSIMA-Team

FIGURE A5. CONTROL LETTER

Notes: Figure shows a screenshot of an example of the letter sent to all students in the control group.

19. Semesterwoche für Test1 Test101: Montag 06.02. bis Sonntag 12.02.2023

1. Meine Lernzeiten während der letzten Woche (in Stunden).

i Nur Zahlen dürfen in diese Felder eingegeben werden.

	Präsenzzeit	Selbstlernzeit	Gesamt
Montag, 30.01.23			0
Dienstag, 31.01.23			0
Mittwoch, 01.02.23			0
Donnerstag, 02.02.23			0
Freitag, 03.02.23			0
Samstag, 04.02.23			0
Sonntag, 05.02.23			0
Gesamt	0	0	0

2. Meine Lernzeitziele für diese Woche (in Stunden).

i Nur Zahlen dürfen in diese Felder eingegeben werden.

	Präsenzzeit	Selbstlernzeit	Gesamt
Montag, 06.02.23			0
Dienstag, 07.02.23			0
Mittwoch, 08.02.23			0
Donnerstag, 09.02.23			0
Freitag, 10.02.23			0
Samstag, 11.02.23			0
Sonntag, 12.02.23			0
Gesamt	0	0	0

FIGURE A6. LEARNING JOURNAL

Notes: The figure shows a screenshot of week 19 of the semester, as an example.

2. Meine Lernzeitziele für diese Woche (in Stunden).

Erinnerung: Sie haben eine Zielvereinbarung abgeschlossen, dass Sie sich den im Lern-Journal formulierten Studier- bzw. Lernzeiten verpflichtet fühlen und diese einhalten werden.

! Nur Zahlen dürfen in diese Felder eingegeben werden.

	Präsenzzeit	Selbstlernzeit	Gesamt
Montag, 10.10.22	5		5
Dienstag, 11.10.22	4	3	7
Mittwoch, 12.10.22	2	5	7
Donnerstag, 13.10.22	6		6
Freitag, 14.10.22		5	5
Samstag, 15.10.22		2	2
Sonntag, 16.10.22			0
Gesamt	17	15	32

FIGURE A7. LEARNING JOURNAL VIEW FOR THE JOURNAL AND COMMITMENT GROUP

Notes: The figure shows a screenshot of the learning journal with a reminder of the commitment device for a student in the second treatment group, who signed the commitment device

***Zielvereinbarung für ein erfolgreiches Studium**

Mit dem Abschluss dieser Zielvereinbarung erkläre ich, dass ich mich den im Lern-Journal formulierten Studier- bzw. Lernzeiten verpflichtet fühle, und diese zum Zweck eines erfolgreichen Studiums einhalten werde.

Diese Vereinbarung wird getroffen zwischen **Test19 Test119**, und dem **Kompetenzzentrum KoSIMA** an der **TH Nürnberg**.

Sie unterzeichnen die Zielvereinbarung digital, indem Sie das entsprechende Kästchen anklicken und Ihren Vor- und Nachnamen eingeben.

! Kommentieren wenn eine Antwort gewählt wird
! Bitte wählen Sie maximal eine Antwort.
! Sowohl das Kästchen muss angekreuzt als auch der Name eingegeben werden.

☒ Ja, Zielvereinbarung digital abschließen für:

☐ Nein, Zielvereinbarung nicht digital abschließen für:

FIGURE A8. COMMITMENT DEVICE

Notes: The figure shows a screenshot of the Commitment device offered to the second treatment group.